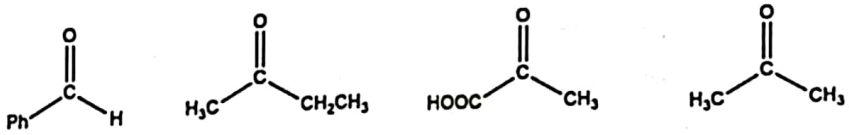
Department of Chemistry
B.Sc. (Hons.) Part-IV, Final Examination 2022
Course no: 432, Course Name: Advanced Stereochemistry
Unit 1/2; Marks = 35; Time 2.5 hrs.

Answer any FIVE of the following questions

1. a. Explain the terms enantiotopic ligand, diastereotopic ligand, enantiotopic faces, and diastereotopic faces with suitable examples. 4

b. Define prostereoisomerism. Assign *Re-face* and *Si-face* of the following compounds: 1+2



2. a. Show the ring flip with equatorial and axial bonds in the chair form of cyclohexane. 1.5

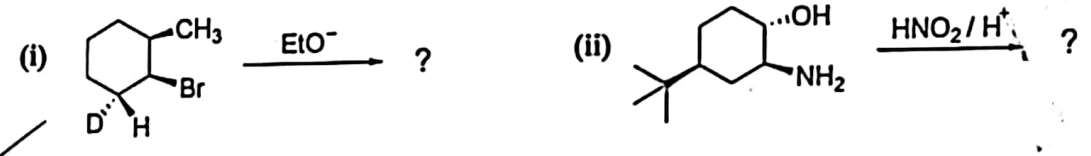
b. Why does cyclobutane possess a slightly bent structure rather than a planner one? 1.5

c. Draw the possible conformations of 1,3 and 1,4 dimethyl cyclohexane mentioning the stability order. 2

d. How can you confirm the conformation of Chlorocyclohexane with ^{1}HNMR? 2

3. a. 2-*tert*-butylcyclohexanol undergoes oxidation reaction to yield 2-*tert*-butylcyclohexanone. 3
Which isomer reacts faster, the *cis*-isomer of 2-*tert*-butylcyclohexanol or the *trans*-isomer? Explain.

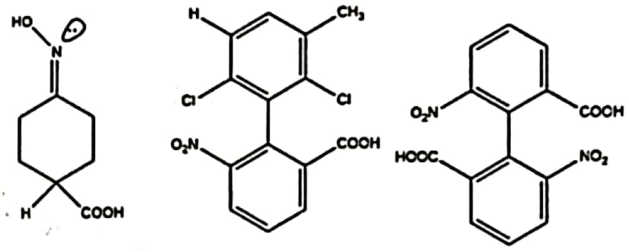
b. Draw all probable products for each of the following reactions and indicate the major product (if there are more than one) and also give explanation in support of your answer. 2+2



4. a. Define atropoisomerism and mention two conditions necessary for biphenyl compounds to exhibit optical activity. 2

b. Discuss allene stereochemistry with proper conditions and structures. 2

c. Find out the absolute configuration of the following compounds: 3

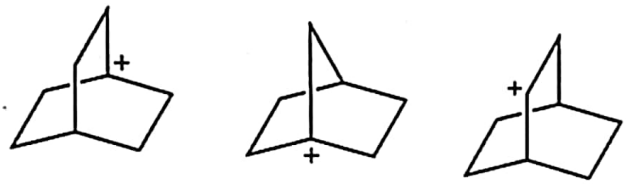


5. a. Explain that Norbornane has a point group C_{2v}. 1

b. Discuss all the conformations of cis and trans-1,4-dimethylcyclohexane, and comment on their optical activity. 3

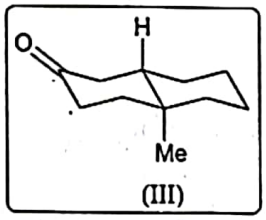
c. Write down the common structural features in steroid molecules. 2

d. Comment on the stability order of the following carbocations based on Bredt's rule: 1



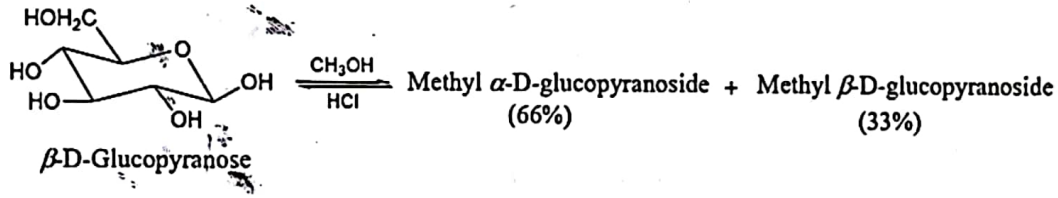
6. a. Point out the difference between optical rotatory dispersion and circular dichroism? 3

b. What is octant rule? The absolute configuration of (+)-enantiomer of *trans*-10-methyl-2-decalone (III) shows positive cotton effect – show how by applying the octant rule. 1+3



(III)

7. a. Consider the following reaction and answer the related question:



β-D-Glucopyranose

(i) Draw the chair conformation of both α and β anomers of methyl D-glucopyranoside. 2

(ii) The formation of the methyl α-D-glucopyranoside is more favored than the β anomer in the above reaction – explain why? 3

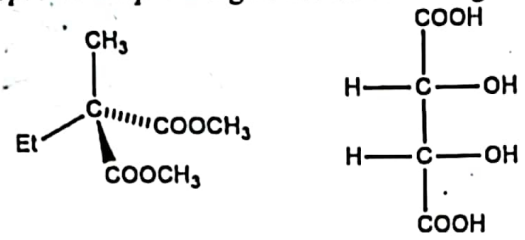
b. What are the main differences between *B*- and *Z*-conformations of DNA? 2

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Department of Chemistry
Jahangirnagar University
B.Sc. Honours Part IV Examination, 2021
Course Title: Advanced Stereochemistry
Course No.: Chem. 432
Full Marks: 35, Time: 2.5 hours

Answer any FIVE questions

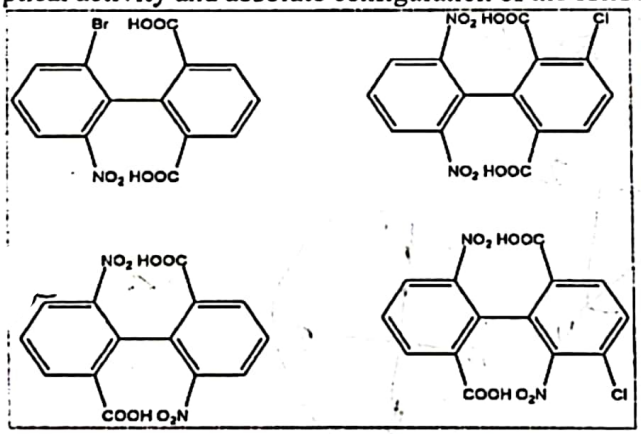
1. a. Chiral reagent like enzyme can distinguish between *pro-R* and *pro-S* hydrogens in a molecule rather than an achiral chemical reagent. Explain with suitable enzymatic reaction mechanism. 3
   b. Explain prostereoisomerism in citric acid cycle. 2
   c. Assign *pro-R* and *pro-S* ligand of the following compounds: 2



2. a. Draw all the possible stereoisomers of dimethyl cyclohexanes. Comment on the optical activity of the different stereoisomers. 2+2
   b. Cyclohexane-1,3-diol exists as a diaxial form. Explain it. 3

3. a. Discuss Edward-Lemiex effect with two suitable explanations. 1.5
   b. Draw the primary structure of amylopectin. 1.5
   c. Why are the cellulose molecules rigid and do not tend to coil into to helical structure? 1
   d. Draw the conformation of the sugar molecule in DNA.
   e. What is β-pleated sheet? Classify it. 2

4. a. Comment on optical activity and absolute configuration of the following compounds. 4

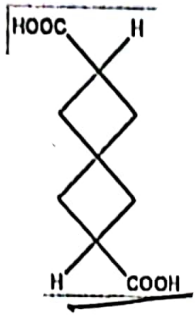


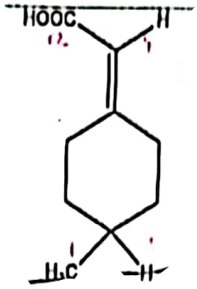
   b. Prove that terphenyl compounds can show both geometrical and optical isomerism. 1.5
   c. Classify 2,2′,6,6′-tetrasubstituted biphenyl compounds. 1.5

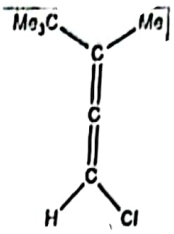
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5. a. Allenes with even number of cumulated double bonds are optically active if the groups on terminal carbons are different, but allenes with odd no of double bonds show geometrical isomerism- Explain this fact with appropriate orbital diagram.

b. Find out the absolute configuration of the following compounds:







c. Outline one synthetic method of allene.

d. Explain optical activity for molecular overcrowding.

6. a. Draw the structures of different stereoisomers of decalin. Comment on the chirality and stability about them.

b. Discuss about the conformation and stereoisomerism of hydrindanes.

c. Define (i) CD (ii) ORD. Describe the role of ORD and CD for determining the configuration of molecule in details.

7. Write short notes on the following: 2+2+3

a. Annomeric effect

b. The obstacle Theory

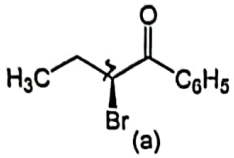
c. Curtin-Hammett principle

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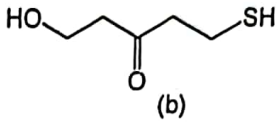
Department of Chemistry
Jahangirnagar University
B.Sc(Hons.) Part-IV, Final Examinations-2020
Course Title: Advanced Stereochemistry
Course No.: Chem-432, Total Marks: 35, Time 2.5 hours

(Answer any five questions)

1. (a) Explain if enantiomers or diastereomers would be formed from the HCN addition reaction of the following ketones(a and b): 2.0

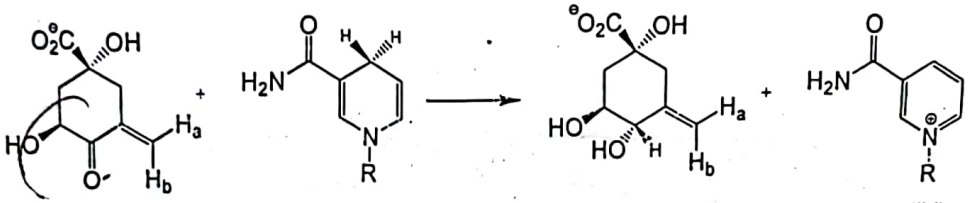


(a)



(b)

(b) Consider the following reaction and answer the related questions:

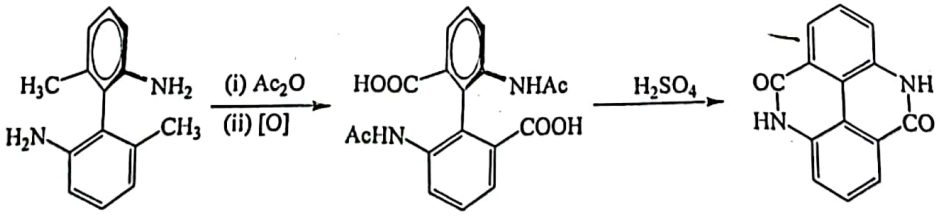


(i) For each of the carbonyl groups in (I) and (IV), explain whether we are looking at the *Re* or *Si* face. 2.0

(ii) Assign the prochiral hydrogens in the compound (II) as H_R *and* H_S. 1.5

(iii) Indicate whether the protons H_a and H_b in the compound (III) are homotopic, enantiotopic or diastereotopic and also give reasons for your answer. 1.5

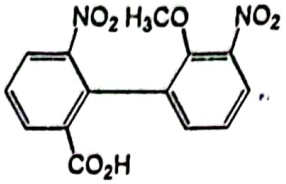
2. (a) Comment on the chirality of the following compounds (1, 2 and 3) giving arguments in each case. Assign *R/S* configuration to the chiral compounds. 3



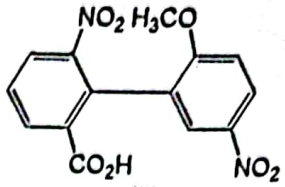
(1) (2) (3)

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(b) Explain why the rate of racemization of (i) is much slower than that of (ii).

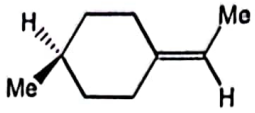


(i)

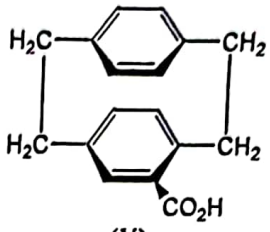


(ii)

(c) Give explanation on the fact that the following compounds (X and Y) are chiral despite not having chiral centers. 2.0



(X)



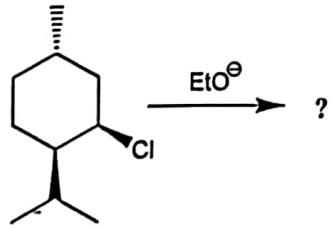
(Y)

3. (a) What types of strain are present in the boat conformation of cyclohexane? – explain by drawing both boat conformation and its corresponding Newman projection. 1.5

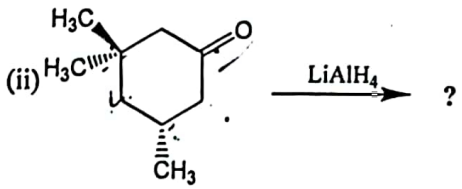
(b) In the chromic acid oxidation of 4-*tert*-butylcyclohexanol, which isomer (*cis* or *trans*) reacts faster and why? 1.5

(c) For the following reactions, draw all probable products and indicate the major product (if there are more than one) and also give explanation in support of your answer. 2+2

(i)



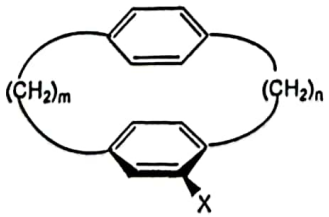
(ii)



4. (a) What are the necessary conditions for showing optical activity of allenes? Explain with a suitable example. 3.0

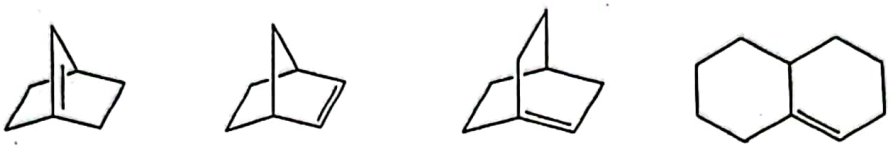
(b) Draw a structure of an optically active spiro compound. Explain the cause of optical activity of such type of compounds. 2.0

(c) Comment on the chirality of the following compound depending on the m, n and x. 2.0



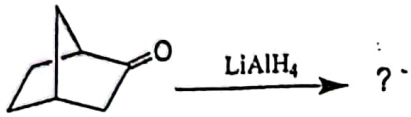
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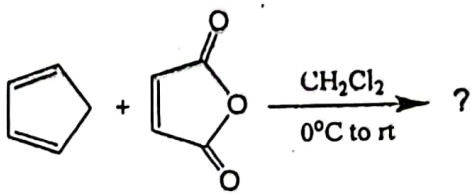
5. (a) Based on the Bredt's rule, comment on the stability of the following bicyclic alkenes' 2.0



(A) (B) (C) (D)

(b) Complete the following reactions and also explain the stereoselectivity of the reactions. 3.0

(i) 

(ii) 

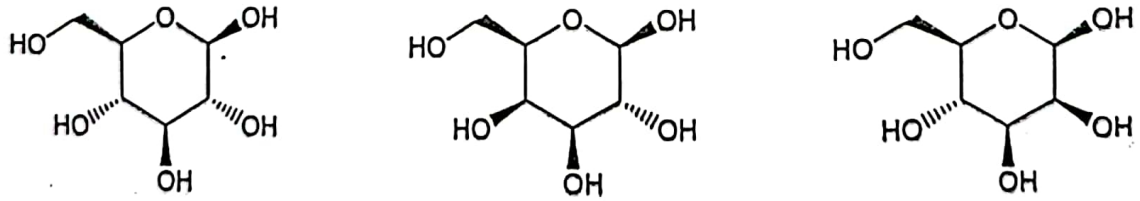
(c) Show the chair conformations of *cis*-Decalin and *trans*-Decalin. Which of these is not able to undergo a ring flip and why? 2.0

6. (a) Define (i) CD (ii) ORD. Describe the role of ORD and CD for determining the configuration of molecule. 3.0

(b) Explain clearly the axial heloketone rule. Mention how the axial haloketone rule can be applied to determine the configuration of trans-1-decalone. 2.0

(c) Explain the configuration of proteins. 2.0

7. (a) Draw the chair conformation of the following monosaccharide molecules (I, II and III). Which monosaccharide is the most stable and why? 3.0



(I) (II) (III)

(b) Explain the role of hydrogen bonds in the secondary structure of protein. 2.0

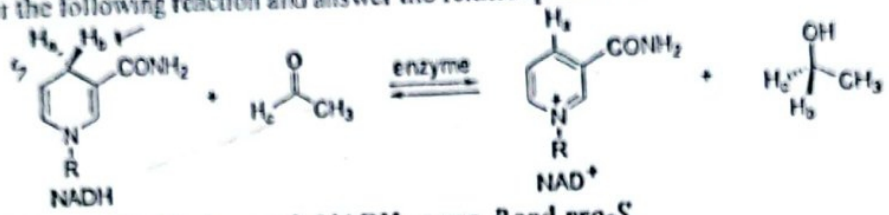
(c) Show the structures of *syn*- and *anti*-adenosine and indicate the glycosyl bond. 2.0

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part IV Examination 2018
Course Title: Advanced Stereochemistry;
Course No. Chem 432, Full Marks: 3[illegible], Time: 2.5 hrs.

(Answer any four questions)

1. (a) Consider the following reaction and answer the related questions:



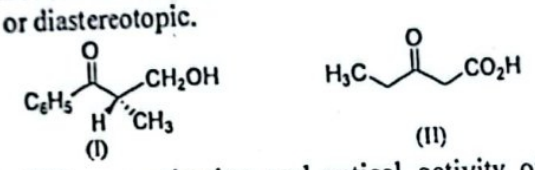
(i) Label the prochiral hydrogens in NADH as *pro-R* and *pro-S*. 1.5

(ii) Depict the top face in acetaldehyde as *Re* or *Si* 0.75

(iii) Indicate the methylene hydrogens in ethanol as enantiotopic or diastereotopic. Give your reasoning. 1.5

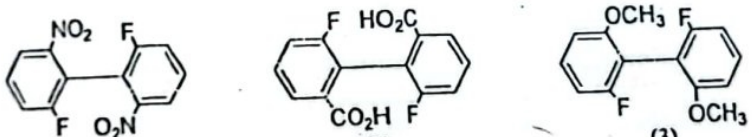
(b) Show the hydride attack from $LiAlH_4$ from both the *Re* and *Si* faces of 2-butanone 2

(c) Using HCN addition reactions identify the faces of each of the following compounds as homotopic, enantiotopic or diastereotopic. 3



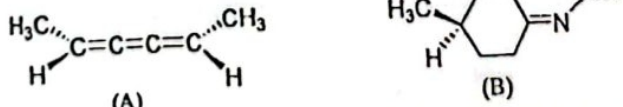
(I) (II)

2. (a) Comment on the resolvability, racemization and optical activity of the following biphenyl compounds: 1.5×3



(1) (2) (3)

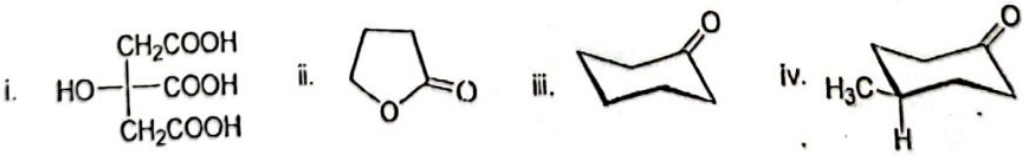
(b) Which of the following compounds exhibits optical activity and which does not? Explain your reasoning and assign *R/S* configuration to the chiral compound. 3+1.25



(A) (B)

3. (a) Define homotopic, enantiotopic and diastereotopic ligands with suitable examples. 3.0

(b) Locate enantiotopic and diasteretopic groups or faces in the following giving arguments. 2.75



(c) Discuss the symmetery criteria of homotopic and enantiotopic ligands and faces with examples. 3

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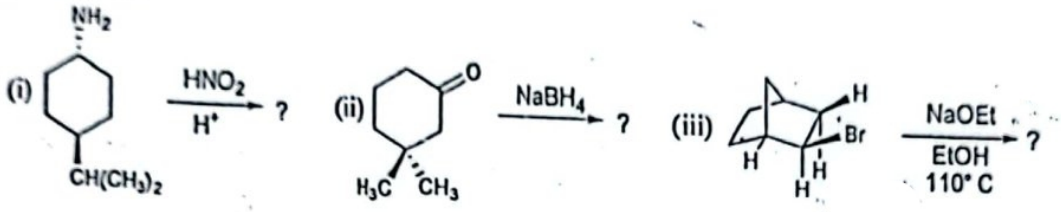
4. (a) Show the most stable conformational isomer of *cis*-1,3-dimethylcyclobutane and justify your answer. 1.5

(b) Predict the position of the equilibrium and major conformer in the following case with proper reasoning. 2.25



(c) With respect to reactivity of cyclohexanes, briefly explain the concept of steric hindrance and steric assistance giving suitable examples and energy diagrams. 2.5×2

5. (a) Predict the major product in each of the following reactions and give explanation in favor of your answer. 2×3



(b) Show the chair conformations of *cis*- and *trans*-decalin and make a comparison between them on basis of their stability and chirality. 2.75

6. (a) Draw the partial conformational structures of cellulose and starch and highlight the major differences between them 3

(b) How do you account for the fact that the axial isomer has a preference over equatorial isomer in the following case? 2.5



(c) What interactions stabilize the tertiary structure of a protein? 1.75

(d) What are the three important conformations of DNA and which one is the most stable? 1.5

7. (a) Differentiate between conformation and configuration. 2.75

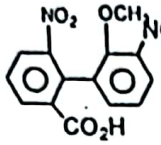
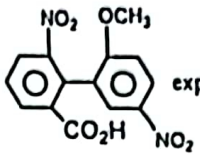
(b) Describe the α-Helix structure of protein. 3

(c) Show the conformation of starch and cellulose. 3

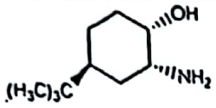
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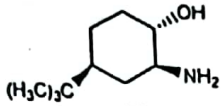
Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-IV Final Examination 2017
Course Title : Advanced Stereochemistry
Course # Chem. 432, Total Marks: 35, Time: 2.5 hours

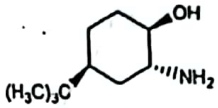
[Answer any four questions]

1. (a) 'The presence of a chiral carbon is not always essential for a compound to exhibit optical activity'---- explain. 3
(b) Give three suitable examples as evidence of the obstacle theory. 3
(c) The rate of racemization of  is much slower than that of  explain. 2.75

2. (a) Draw the chair form of the following compounds: 1.5


(I)


(II)


(III)

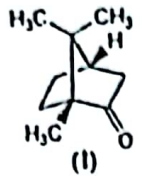
(i) Which one of these compounds is least stable? Give explanation. 0.5
(ii) Are all of these compounds formed the same product by the reaction with nitrous acid? Give chemical reactions and explanation in support of your answer. 3
(iii) Is there any function of the *tert*-butyl group that present in each of the molecules? 0.75
(b) After a little bit puckering of the planar structure of cyclobutane forms a non-planar structure.
(i) Draw the non-planar structure of cyclobutane and its corresponding Newman projection formula and answer the following questions: 1
(ii) Are the C-H bonds in the Newman projection formula are perfectly eclipsed or not? 0.5
(iii) Is the angle strain increased or decreased after the puckering of the planar structure of cyclobutane? 0.5
(iv) Which one of the structures (planar or puckering) of cyclobutane is more stable? Give explanation in support of your answer. 1

3. (a) Draw the conformational (partial) structure of cellulose and give satisfactory answer to the following questions:
(i) What are the monosaccharide units of cellulose and what type of linkage present between the monosaccharide units? 1.25
(ii) Why are the cellulose molecules rigid and do not tend to coil into the helical structure? 0.5
(iii) Why is the cellulose molecule insoluble in water although it has a lot of hydroxyl groups? 1.5
(b) Draw the chair form of α-D-glucopyranose and β–D- glucopyranose and answer the following questions with explanation on the basis of their structures: 1.0
(i) Which one of the isomers is more stable? 1.5
(ii) Indicate the anomeric carbon atom in each of the isomers. 1.
(iii) If the hydrogen atom of the hydroxyl group bonded to the anomeric carbon atom in each of the isomer is replaced by the methyl group then which one of the resulting isomers will be more stable? Give explanation in support of your answer. 0.5 1.5

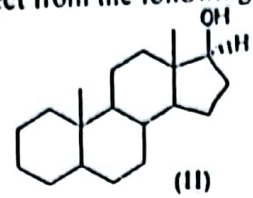
4. (a) Define the term Circular Dichroism (CD). Briefly discuss its basic principle. 1+2.5=3.5
(b) What are the basic differences between the ORD and CD spectroscopy? 1.75
(c) What is ORD curve? How does an ORD curve represent? 1
(d) There are two main types of ORD curves
(i) What are the names of these two main types of ORD curves? 0.5
(ii) Draw these two ORD curves? 1

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(c) What type of ORD curve would you expect from the following two compounds?



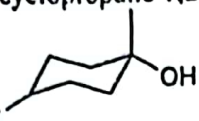
(I)



(II)

5. (a) Explain with example fundamental basis of curtin-Hammett principle with regard to the proportioner of more than one product in a reaction. 4

(b) Show the stereoisomers of cyclopropane 1,2-dicarboxylic acid. 2

(c) How will you synthesize 

from tert-butyl cyclohexanone as the major product? 2.75

6. (a) Complete the following reactions and identify as homotopic and enantiotopic faces : 2

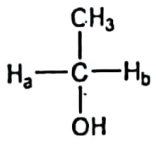
(i) $HCHO + CH_3MgI \dashrightarrow ?$

(ii) $CH_3CHO + DCN \longrightarrow ?$

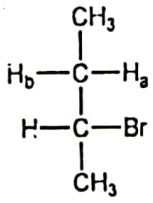
(b) How are enantiotopic and diastercotopic faces named Using Chan-Ingold-Prelog rule? Show with examples. 2.75

(c) Discuss how enantiotopic faces can be distinguished using enzymatic oxidation or reduction. Show with appropriate examples. 4

7. (a) Consider the two compounds (I & II) and give satisfactory answer to the following questions with appropriate explanation:



(I)



(II)

(i) Are the two methylene hydrogen atoms (H_a & H_b) of the two compounds I and II homo- topic of heterotopic in nature? 1.75

(ii) Which of the two methylene hydrogen atoms (H_a & H_b) of the two compouns I&II are enantiotopic? 2

(iii) Which of the two methylene hydrogen atoms (H_a & H_b) of the two compounds I and II are diastereotopic? 2

(iv) Assign the two enantiotopic methylene hydrogen atoms (H_a & H_b) as *Pro-S-H* and *Pro-R-H*. 1.5

(b) Does the term prostereogenic center always mean a prochiral center? Explain with suitable examples. 1.5

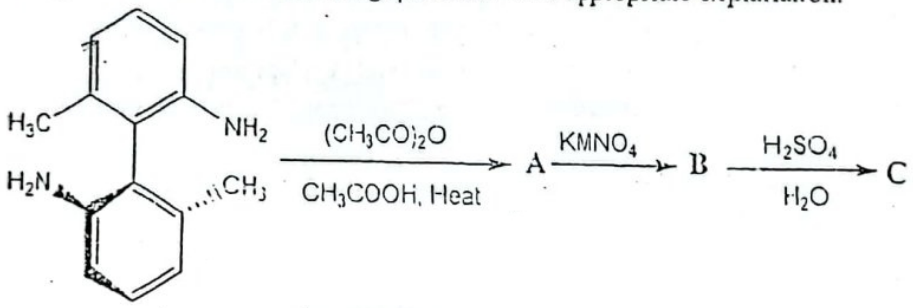
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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) 4th Yr. Final Examination, 2016
Course Title: Advanced Stereochemistry
Course: Chem. 452
Time: 2.5 hours, Full Marks: 35

[Answer any FOUR questions]

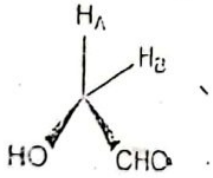
1. (a) Write down the name of the smallest member of cycloalkane. 0.5
(b) Draw the planar structure and its corresponding Newman projection formula of the smallest cycloalkane and answer the following questions: 1
(i) Is the cycloalkane free from angle strain? 0.5
(ii) Is there any eclipsed C-H bonds? 0.5
(iii) What kind of strain present between the C-H bonds of the cycloalkane? 0.5
(c) Draw the chair form and it's corresponding Newman projection formula of cyclohexane and cyclohexanone and answer the following questions: 2
(i) Is there any eclipsed hydrogen atom in the cyclohexane molecule? 0.5
(ii) Is there any eclipsed hydrogen atom in the cyclohexanone molecule? 0.5
(iii) Is the cyclohexanone molecule free from angle strain? 0.5
(iv) Which one of the compounds is more stable? Give explanation in support of your answer. 2.25

2. (a) What are the necessary conditions for biphenyl compounds to exhibit optical activity? Explain with suitable examples. 2.75
(b) Give structures of the unknown compounds A-C in the following series of reactions and give answer to the following questions with appropriate explanation: 1.5



(i) Which of these products are chiral? 1
(ii) Which of the products is achiral? 0.75
(iii) Assign the *R*/*S* configuration of the chiral products that form in the above reaction. 2
(iv) Is the starting compound chiral or achiral? 0.75

3. (a) Define pro-R and pro-S. Identify the ligands H_A and H_B as pro-R and pro-S in the following compound. 2.75



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(b) What are homotopic and heterotopic atoms? Are the 2H's of CH_2ClF homotopic? 2

(c) Explain briefly a relation between elements of symmetry and optical activity. 2

(d) How enantiotopic faces can be distinguished using enzymatic oxidation or reduction? Show with appropriate examples 2

4. (a) Differentiate between conformation and configuration. 2

(b) Describe the α-Helix structure of protein. 2.75

(c) Show the conformations of starch and cellulose. 2

(d) What is 'anomeric effect'? Which anomer is more stable between α and β-anomer? 2

5. (a) Explain the phenomenon of "Circular dichroism". 2.75

(b) What is axial haloketone rule? Explain this rule and provide its applications. 1+2

(c) What is "Cotton effect"? How does cotton effect help to analyse the configuration and conformation of cyclic ketones? 1+2

6. (a) Draw the planar structure of decalin with the numbering of all carbon atoms and answer the following questions: 1

(i) Is the compound a fused or a bridged ring bicyclic compound? 0.5

(ii) Write another two names of the compound. 0.5

(b) Decalin exists in two stereoisomeric forms:

(i) What are the names of these two stereoisomeric forms of decalin? 1

(ii) Draw the planar structures and the corresponding chair conformations of the two stereoisomeric forms. 2

(iii) Which one of the stereoisomeric forms of decalin is more stable? Give explanation in support of your answer. 2.5

(c) "The elimination of HBr from bromocamphor in the presence of a base is not possible" - Justify the statement on the basis of the Bredt's rule. 1.25

7. (a) Define the terms "ansa" and "cyclophane" compounds? Cite examples. 1.25

(b) What is the basic difference between "ansa" and "cyclophane" compounds? 0.75

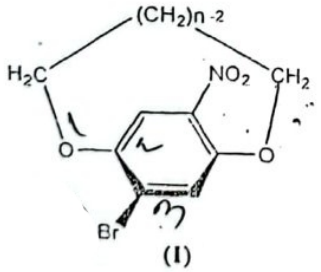
(c) What are the necessary conditions for "ansa" compound to exhibit optical activity? Explain with suitable examples. 2.5

(d) Define the term "pilot atom" and give answer to the following questions:

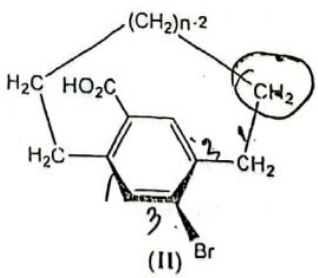
(i) How is a pilot atom marked? 0.5

(ii) How would you select the pilot atom? 0.75

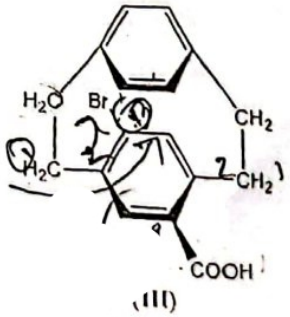
(e) Assign the R/S configuration of the following compounds: 3



(I)



(II)

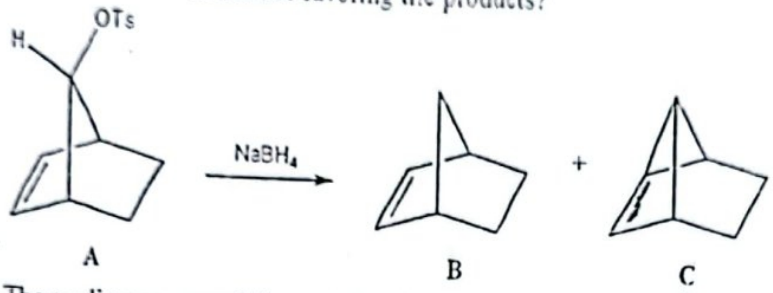


(III)

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-IV Final Examination 2015
Course Title: Advanced Stereochemistry
Course # Chem. 432, Total Marks: 35, Time: 2.5 hours

[Answer any four questions]

1. (a) The solvolysis of compound A with $NaBH_4$ affords compound B and C. How do you explain the reaction mode favoring the products? 2.5



A B C

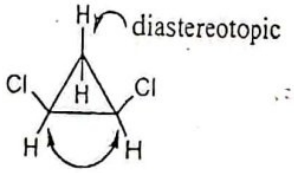
(b) The cyclic compound shown below is a bridged compound. Answer the following questions regarding the compound. 3



(i) The above type of compound is extremely unreactive in S_N2 reaction-explain.
(ii) The compound is more unreactive even in S_N1 reaction compared to acyclic compound.

(c) Using symmetry considerations, show two H,s of CH_2ClF are homotopic or not? 2

(d) The hydrogen atoms on C-1 and C-2 in cis -1,2-dichloropropane have been labeled as shown below. Give arguments. 1.25



enantiotopic

2. (a) Write the conformations of 1,3-dimethylcyclohexane. Discuss the chirality and stability of *cis* and *trans*-isomers. 2.5+1.5

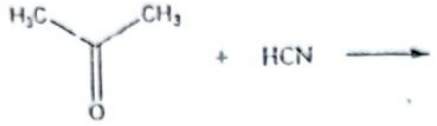
(b) Discuss the geometrical isomerism of 1-bromo-2-chlorocyclobutane. Give stereodescriptors to different stereocenters. 2.0+1.

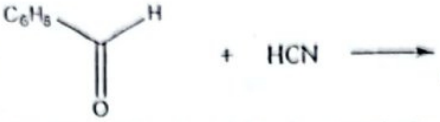
(c) (R)-camphor is reduced with lithium aluminium hydride to give a pair of alcohols. Give nomenclature to each stereocenter in these alcohols. How these alcohols are stereochemically related? 1.75 2.75

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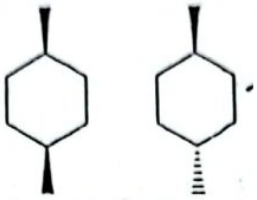
3. (a) Define Pro-R and Pro-S center with suitable example. 2

(b) Complete the following reactions and identify them as homotopic- or enantiotopic-face. 2.5

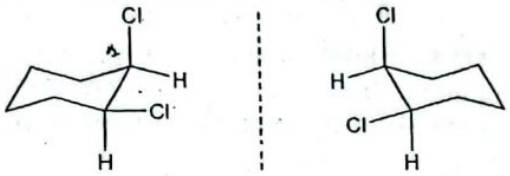
(i) 

(ii) 

(c) Which of the following C_8H_{16} isomers is thermodynamically most stable and why? 2.25



(d) The drawing below shows that cis-1,2-dichlorocyclohexane is chiral. Efforts to resolve this compound fail-Explain why? 1.75



4. (a) What are the necessary conditions for showing optical activity by the biphenyls? Explain with suitable examples. 2

(b) Explain the terms chiral plane and chiral axis with suitable examples. 1.5+1.5

(c) Write down the synthetic plan of optically active allenes. 2

(d) What is a pilot atom? Draw the structure of *trans* –cyclooctene and discuss its stereochemistry. 1.75

5. (a) What is the difference between alpha –helix and beta-sheet protein conformations? 1.5

(b) What is protein denaturation? Are there any changes in the primary structure when a protein is denatured. 1.5

(c) Draw and discuss the tertiary structure of proteins based on their conformations. 2.75

(d) How does cellulose differ from amylopeptine conformationally? 1

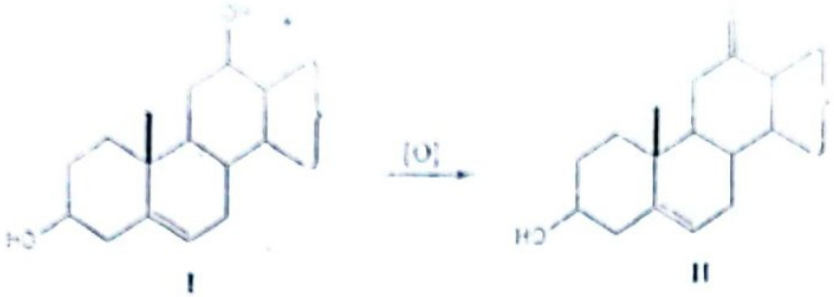
(e) The pyranose ring system of hexose prefers the axial orientation instead of the less hindered equatorial orientation –explain it by means of anomeric effect. 2

6. (a) What do you mean by chirooptical properties? Explain it with suitable examples. 1

(b) Define ORD and CD. Compare and contrast between them in details with examples with the appropriate curves. 2+2=4

(c) The following dihydroxy compound, I upon oxidation gives the hydroxyl ketone, II. How could you prove that the only one hydroxylgroup can be oxidized to give II? 1.75

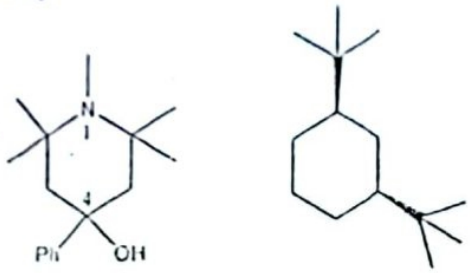
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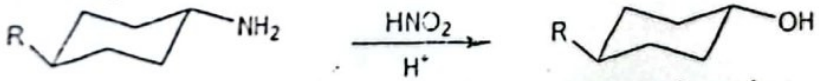
I II

(d) What is octant rule? Explain the rule and provide its application 2

7. (a) Draw all boat and chair conformations for the following molecules and predict their conformational stability. 2



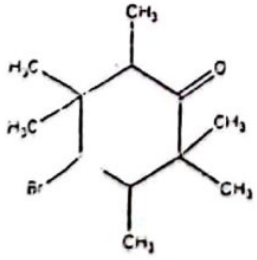
(b) The following reaction gives the equatorial alcohol as a sole product from the corresponding amine. Give a mechanism, which describes the stereochemical features of the reaction nicely in favor of the formation of the equatorial alcohol as a sole product. 2



sole product

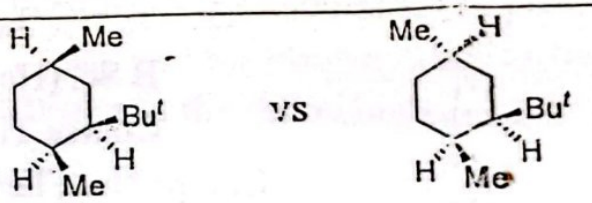
(c) What are inositols? Draw the conformational structure of myoinositol. Give three important biological importance of myoinositol. 2.75

(d) Examine the following compound and write down the number of all the possible stereoisomers having this constitution. 2



Tutorial Examination-02; Chem 432; Marks : 10; Time: 45 Minutes

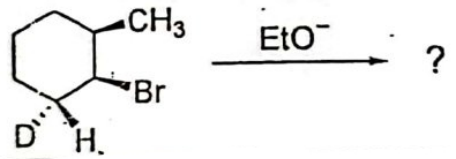
Which of the two isomers of tri-substituted cyclohexane (I & II) is the most stable? Explain your reasoning. 2

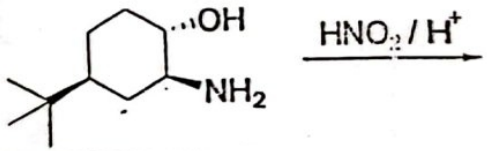


(I) (II)

2. In the saponification of ethyl 4-*t*-butylcyclohexane-1-carboxylate, the *trans* isomer reacts faster than *cis* isomer – explain why [Be sure to draw the energy level diagram.] 4

3. For the following reactions draw all probable products and indicate the major product (if there are more than one) and also give explanation in support of your answer. 2+2

(i) 

(ii) 

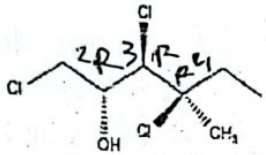
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B. Sc. (Hons.) – 4th year: Tutorial – 1 Time: 40 min

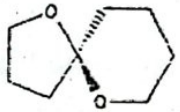
(a) Define pro-chirality with an example. Pick out the molecules which are pro-chiral from the following: 1

$(CH_3)_2CHOH$ $CH_3CH_2CH_2CH_3$ CH_3CH_2Br $CH_2(Cl)(Cl)$ CH3CHO

(b) Label each asymmetric carbon with R or S. 2



(2R,3R,4R)

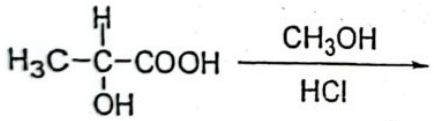


S

(c) Draw the following compounds: (i) (R)-2-bromo-1-butanol; (ii) (2R,3S)-2-chloro-3-pentanol 2

(d) Draw Fischer projection formulas for all stereoisomer's of 2,4-dibromohexane, giving stereochemical details for each structure. 3

(e) Esterification of (+)-lactic acid with methyl alcohol gives (-)-methyl lactate. 2



(+)-Lactic cid

Answer the following questions based on the above statement.

i. Complete the reaction

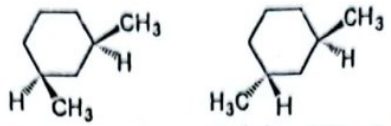
ii. has the configuration changed?

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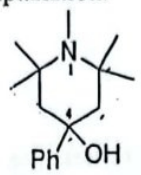
Chem - 432 Tutorial 2

Marks: 10 Time: 40 min

1. (a) Write all chair conformations for the following 1,3-dimethyl cyclohexanes. which one is the most stable conformer and why? 2

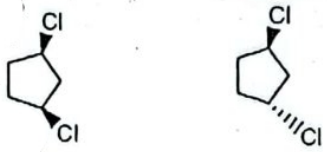


(b) Draw the stable conformer/conformers for the following molecule. Indicate the most stable one with a proper explanation. 2

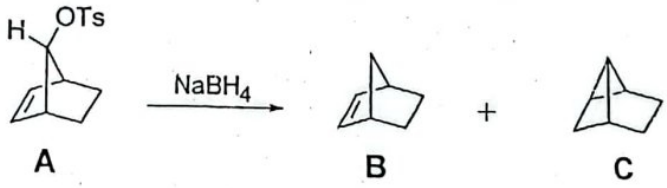


1

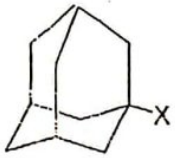
(c) Which of the following compound is not optically active and why? 1



2. (a) The solvolysis of compound **A** with $NaBH_4$ affords compound **B** and **C**. How do you explain the reaction mode favouring the products? 2



(b) The cyclic compound shown below is a bridged compound. Answer the following questions regarding the compound



i) The above type of compound is extremely unreactive in S_N2 reactions-explain. 1.5

ii) The compound is more unreactive even in S_N1 reaction compared to noncyclic compound. 1.5

CamScanner

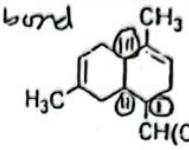
Chem -432

Tutorial I

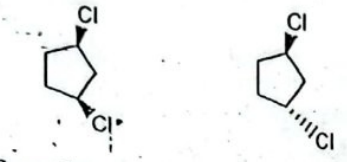
Marks: 10 Time: 40 min

1. (a) Name all types of strains with due reasons for cyclohexane ring feels because of its conformational changes. i. steric strain due to flagpole hydrogen. ii. Torsional strain due to the eclipsed bond 1

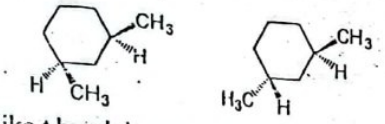
(b) How many chiral carbons are in the compound ?-cadinene? 1



(c) Which of the following compound is not optically active and why? 2



2. (a) Write all chair conformations for the following 1,3-dimethyl cyclohexanes. Which one is the most stable conformer and why? 2



(b) The bulky group like *t*-butyl does not tolerate the conformational flip into an axial orientation in the cyclohexane ring system- explain. 1

(c) Write the products and give an account for the different conformational products' ratio (A: B, a:e) of the following reactions scheme. 2



(d) Show the biological importance of chirality with a suitable example. 1